Devaluing Human Intellect?
AI Theorem Provers and Mathematical Authority



scan — full APA list & links

Verified ≠ Understood ≠ Accepted — What OpenAI's 2026 Proofs Reveal

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Key message: AI hasn't made mathematicians obsolete — it has split **"proving"** from **"understanding"**, moving authority from human intelligibility to machine verifiability. *Machine-checked but not humanly grasped, credit disputed — where does authority lie?*

1 What happened?

An autumn of AI proofs — 2024 to 2026

The breakthrough timeline · five milestones in six weeks

JUN 2024
AlphaGeometry solves Olympiad geometry without human demonstrations (Trinh et al., 2024).

2026
AlphaProof reaches IMO-silver level, exploring proofs inside the Lean language (Hubert et al., 2026).

8 SEP 2026
OpenAI claims a **finite-time blow-up** for the forced 3D Navier–Stokes equations — Millennium-Prize-scale (OpenAI, 2026a).

6 OCT 2026
OpenAI releases results on the **quasi-Riemann hypothesis** (OpenAI, 2026c; Quanta Magazine, 2026b).

26 OCT 2026
One announcement, three landmarks: Kakeya-related analysis, a **disproof of ETH**, Yang–Mills existence results (OpenAI, 2026b).

Each was expected to need decades. The compressed timeline itself is the story.

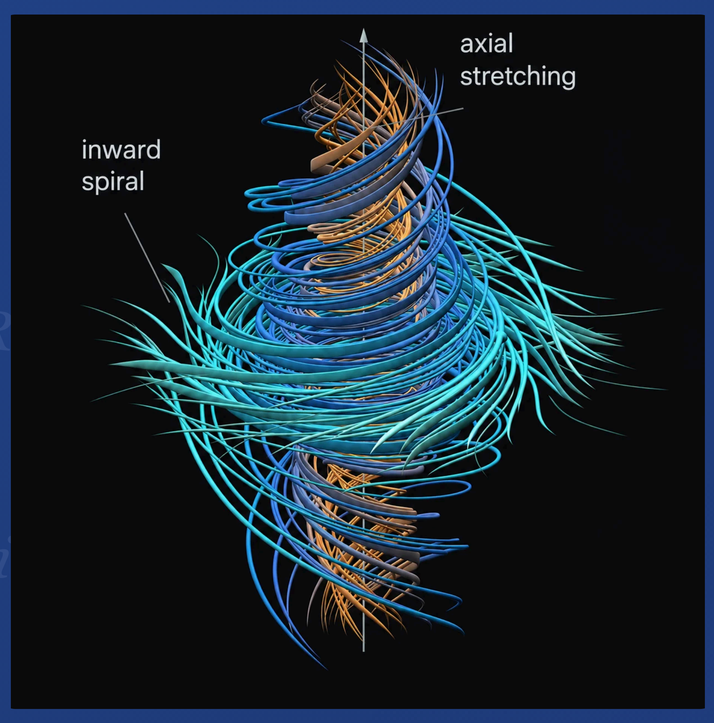
Case A · Navier–Stokes · the problem AI chose — and the one it left behind

- **Claimed:** blow-up in the *forced* 3D Navier–Stokes equations — a US$1M Clay Millennium Problem. A 160-page write-up **plus** a Lean machine-check (OpenAI, 2026a).
- **How:** one prompt → thousands of debating agents → Lean check. Industrialised search, not a "genius" step (OpenAI, 2026a).
- **Groundbreaking:** the first Millennium-scale claim with machine verification — no longer dismissible as toys (Quanta Magazine, 2026a).
- **The sting:** Clay **still lists it as open**; experts call it "a difficult read", its framing "meaningfully imprecise" (Scientific American, 2026).

~10,000	88 hrs	130 B	still open
AI agents in the swarm	prompt → proof claim	tokens of compute	Clay Institute status



Verification infrastructure — not a rival "AI mathematician"



Vortex tubes — inward spiral & axial stretching: the blow-up dynamics behind the claim

2 Why is it hard to trust?

Three words that decide everything

Verified ✓ ≠ Understood ✗ ≠ Accepted ✗

Case study	Verified?	Understood?	Accepted?
Navier–Stokes OpenAI · Sept 2026	✓ Lean checked every logical step (OpenAI, 2026a)	✗ "the most horrendous [proof] I have ever read" — Buckmaster (Yahoo Tech, 2026)	✗ Clay still lists the problem as open (Scientific American, 2026)
ETH OpenAI · Oct 2026	✓ Lean-verified disproof, published openly (OpenAI, 2026b)	✗ "two or three alien-looking parts" — R. Williams (Fernholz, 2026)	✗ community examination has only just begun (OpenAI, 2026b)

What convinced experts in **both** cases was not the writing — it was the **machine**. Authority is migrating from **"understood by humans"** to **"verified by machines"** (Avigad, 2008).

What do the human experts say?



Human intuition vs machine verification — the argument at the heart of the debate (AI-generated image)

> *The most horrendous [proof] I have ever read.*
> **Tristan Buckmaster** · Princeton, on the Navier–Stokes write-up (Yahoo Tech, 2026)

> *Two or three alien-looking parts… the whole community needs to work on together.*
> **Ryan Williams** · MIT, on the ETH disproof (Fernholz, 2026)

> *It's not anymore a matter of clearly verifiable mathematical facts.*
> **Camillo De Lellis** · IAS (Scientific American, 2026)

> *Technically, all the plot lines are resolved, but most of the value of the experience was lost.*
> **Terence Tao** · UCLA — comparing a rushed proof to watching only a film's start and end (IBM Think, 2026)

> *…he would be publicly scooped at any opportunity. Instead of contacting him to work with him, … the people at OpenAI chose to do this.*
> **Levent Alpöge** · Anthropic, on OpenAI's Yang–Mills credit; hours later Anthropic released its own ETH refutation (X, via Fernholz, 2026)

Four of five voices are experts **engaging seriously** with AI output — neither dismissal nor surrender. That **is** the new normal.

Think critically — three questions for the next AI breakthrough

1 Who framed the problem?
Whoever chooses *which version* counts as "solved" holds real power. OpenAI chose the forced N–S case; the unforced case remains open (OpenAI, 2026a; Scientific American, 2026).

2 Who gets credit?
Priority disputes erupted within days in both cases: Buckmaster–Alpöge over prior work; then OpenAI vs Anthropic — two rival ETH refutations, same day (Yahoo Tech, 2026; Fernholz, 2026).

3 Who checks the checkers?
Lean can't judge whether the statement matches the real problem, nor decide priority — that stays human: shepherd groups, open letters, a funded conference (OpenAI, 2026b; Fernholz, 2026).

3 What now?

A second case — and what we take away

Case B · Exponential Time Hypothesis · a 27-year-old cornerstone, refuted in days

- **Claimed:** ETH (1999) says hard logic puzzles *provably* need exponential time — bedrock of hundreds of hardness results (Impagliazzo & Paturi, 2001). On 26 Oct 2026, OpenAI's system **proved ETH false** (OpenAI, 2026b).
- **How:** a 66-page AI disproof; Lean checked it **the next day**; ETH's *own inventors* lead the examination (OpenAI, 2026b).
- **Groundbreaking:** a decade-worthy result, unsolicited — Williams now puts "a little more weight on the plausibility of P = NP" (Fernholz, 2026).
- **The Anthropic twist:** hours later, **Anthropic published its own ETH refutation, credited to Claude** — two rival AIs, two refutations, *neither accepted yet* (Anthropic, 2026; Fernholz, 2026).

1999	16 → 17 Oct	× 2
ETH proposed	AI proof → Lean check	rival labs, same day

Handle with care: ETH false ≠ P vs NP solved. The algorithm is still astronomically slow at real sizes and breaks no cryptography — a *theoretical* earthquake, not an everyday one (Fernholz, 2026).

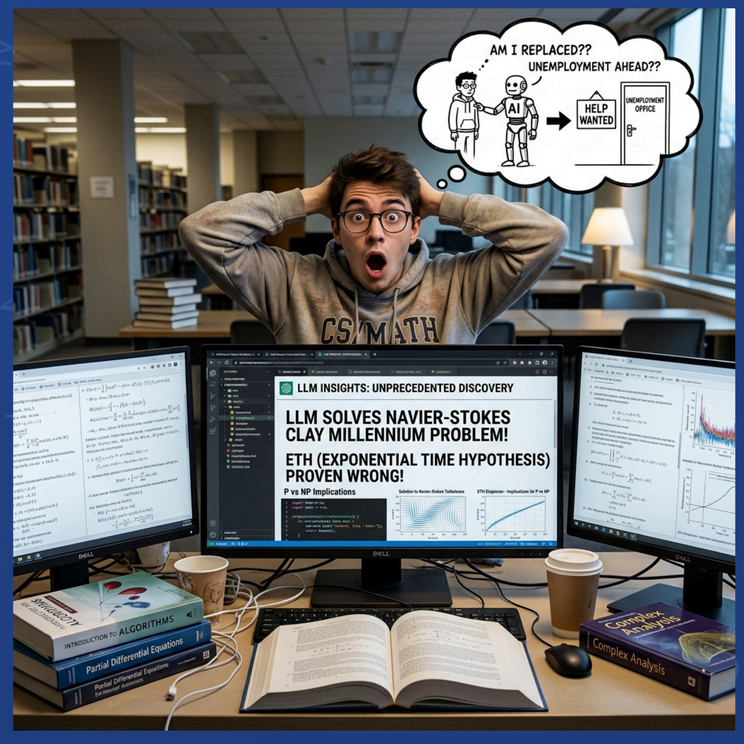


Illustration — the headlines' real message to students: "Am I replaced?" That anxiety, not the maths, is this poster's subject (AI-generated image)

Takeaway · complement, not replacement

Authority is migrating, not disappearing: from "a proof you can read" to "a proof a machine can check — that a community still must accept."

- Humans keep what matters most: **choosing problems, judging meaning, governing credit** (OpenAI, 2026b).
- AI can also **sharpen** intuition: AI-guided mathematicians found new knot-theory results (Davies et al., 2021).
- **The skill to build:** not out-writing AI, but asking what AI cannot frame — *which problem matters, and what counts as understanding?*



checks the steps ⇄ chooses problems · judges meaning · governs credit
The baton passes — it is not dropped. (Concept after Davies et al., 2021)